Aakrit Sharma

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PROFESSIONAL SUMMARY

Highly accomplished Computer Engineering undergraduate specializing in AI Engineering, Large Language Model (LLM) systems, agentic workflows, and scalable inference infrastructure. Proven ability to design, develop, and deploy comprehensive AI solutions, including advanced retrieval systems, ranking engines, and production-oriented AI applications. Adept at applying deep learning and machine learning principles, with strong foundations in NLP, distributed systems, and backend engineering using Python and PyTorch. Eager to contribute expertise in R&D, solutions architecture, and ethical AI integration to drive strategic business benefits and advise on technology innovation.

EDUCATION

[University Name - Not Provided] [Dates - Not Provided]

Computer Engineering Undergraduate Studies

EXPERIENCE

Outstyl Aug 2025 – Sep 2026

Machine Learning Intern

- Developed transformer-based recommendation systems using BERT4Rec and SASRec for predictive user behavior modeling and ranking optimization.
- Built scalable PyTorch training pipelines for sequential recommendation and representation learning workflows.
- Improved recommendation quality through temporal embeddings, transformer ranking architectures, and evaluation pipelines.

PROJECTS

AI Candidate Ranking Engine *Embeddings, LLMs, FastAPI, Docker, Ranking Systems*

- Designed a large-scale candidate evaluation pipeline processing 100,000+ resumes using embeddings, ranking signals, and cross-attention architectures.
- Built precomputation, indexing, and retrieval workflows enabling low-latency candidate search and ranking.
- Integrated local LLM-based reasoning modules to generate explainable hiring recommendations and ranking justifications.
- Containerized the inference stack using Docker for reproducible deployment and scalable serving.

Persistent Context Memory Store *MCP, Agentic AI, Context Engineering, Retrieval Systems*

- Built a local-first memory system enabling AI agents and IDE assistants to share persistent context through the Model Context Protocol (MCP).
- Developed automated memory-generation pipelines that convert conversations and project artifacts into structured Markdown-based knowledge repositories.
- Implemented semantic retrieval, memory ranking, and context selection mechanisms for long-horizon agent workflows.
- Maintained a unified global memory layer across development environments, reducing context fragmentation between AI tools.

DualSpeechLM Optimization Pipeline *LLMs, Multimodal AI, LoRA, Quantization, PyTorch*

- Designed adaptive token-routing and context-compression modules for efficient multimodal speech-language model inference.
- Integrated Phi-3.5-mini-Instruct using 4-bit NF4 quantization and LoRA fine-tuning, significantly reducing memory requirements while preserving performance.
- Built GRU-based dialogue-state tracking, semantic importance scoring, and prosody-aware compression pipelines for long-context conversations.
- Developed auxiliary objectives for dialogue-act classification, semantic alignment, and emotion-aware response generation.

SKILLS

Technologies AI Strategy & Solutions, Machine Learning (ML), Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Embeddings, Agentic AI, Multimodal AI, Retrieval Systems, Ranking Systems, Context Engineering, Scalable Inference, Model Deployment, Python, PyTorch, FastAPI (REST API Development), Docker, Basic Algorithms, Object-Oriented Design Principles, Functional Design Principles, Solutions Architecture, Distributed Systems, Backend Engineering, NoSQL Database Design, RDBMS Design & Optimizations, Program Leadership, Ethical AI, Innovation Accelerators

ACHIEVEMENTS

Adobe India Hackathon

Top 100 / 250,000+ Teams – for building an AI-powered document intelligence platform.

CTRL+SPACE Hackathon Winner

Winner among 2,500+ participants.

Accelerated Data Science using CUDA and RAPIDS

DSA Problem Solving

Solved 300+ DSA problems across algorithms, graphs, dynamic programming, and advanced data structures.